

STAT400: Midterm Review

Fall 2025

Sampling

- I think it would also be helpful to review how to find an inverse CDF, since I'm still a bit unsure about that.
- Could we also go over approaches for finding the c constant for the envelope in the accept-reject method?
- Lastly, going over the accept-reject method and Monte Carlo sampling would be great!
- What are some ways to come up with an envelope when the uniform does not work?
- What are some good ways to prove that the envelope is valid?
- I am really confused about the method of using envelopes. I don't really understand how to use them, nor why or when we would want to use them.
- I would like you to cover how to create the best 'envelope' for accept-reject method. And please provide an example, I noticed in our notes we don't really have a defined envelope in the examples provided; I just want to see the thinking process on how to create the envelope.
- I would like to request that we go over envelope functions a bit more, and I am struggling a bit to understand specifically what they are and how they are found.

Monte Carlo Integration

- Could we go over a specific example and/or a review of finding the $\text{Var}(g(x))$ using a Monte Carlo approach?
- I was wondering if, when we are using methods 1 & 2 for Monte Carlo Integration, is the uniform distribution the only distribution we can use? In other words, if the uniform distribution isn't a good option, is method 3 our go-to? If not, how would we use other distributions in that way?
- In our review on Tuesday, would you elaborate more on why Monte Carlo integration is superior to approximating an integral via the sum of polygons? I know that we talked about how the estimate produced by Monte Carlo converges towards the true value of the integral as m increases, but I don't quite understand why this is better than increasing the number of polygons, especially because Monte Carlo inherently involves randomness.

Other Questions

- Also, from Homework 3, we had to do integration by parts; could we go over a similar example during Tuesday's review?
- Going over marginal densities and trying a new example that isn't the drive-up and walk-up window one used from notes.
- Could we review some examples using `group_by`, `summarise`, `filter` (etc.) like we did earlier in the semester?
- I was wondering if it would be possible to elaborate on what we can expect from the coding-based questions on the exam.

